

charts can be made for different assumptions about retirement length and the acceptable failure rate). Each diagonal line (technically an “isoquant”) shows the combination of rates of return and standard deviations that will support a given sustainable withdrawal rate.

From this overlay, we determine where the efficient frontier reaches the highest withdrawal rate. The thin parts of the curves identify the entire efficient frontier of asset allocations. The circles identify the points reaching the highest sustainable withdrawal rate. The thick parts of the curves identify the range of portfolio allocations supporting withdrawal rates within 0.1 percentage points of the maximum. Returning to the mean-variance optimization results, we find the asset allocation corresponding to this optimal point. The optimal asset allocation can then be compared to the retiree’s investment constraints and risk tolerance to determine its acceptability. If it is found to be unacceptable, other points on the efficient frontier that support progressively lower withdrawal rates can be investigated until a suitable asset allocation is found.

Exhibit 5.27

Sustainable Withdrawal Rates with 10% Failure for 30-Year Retirement Horizon

